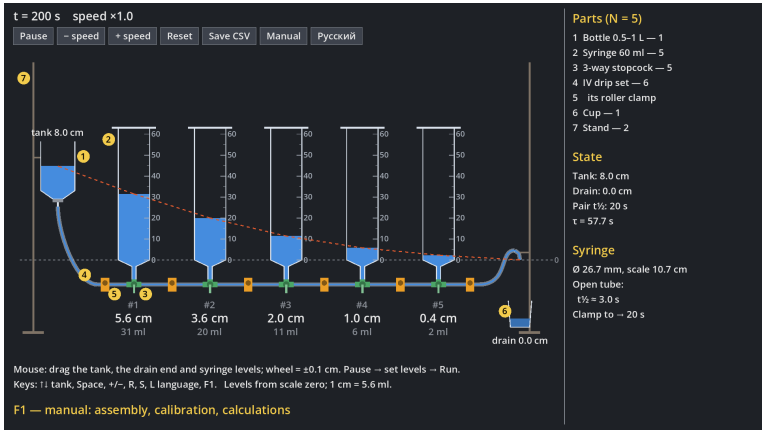
Lukyanov hydraulic integrator from syringes and IV drips

Assembly, calibration and calculation manual



The scheme in the program. F1 opens/closes this manual, ← / → flip pages.

1. What it is and how it works

In 1936 V. S. Lukyanov noticed that water flowing between vessels through narrow tubes obeys the same equation as heat flowing between layers of a body. Split the body into layers, give each layer a vessel, and the water levels follow the layer temperatures — the rig "solves" the heat equation by itself, without formulas.

Each syringe obeys $dh_i/dt = (h_{i-1} - 2h_i + h_{i+1}) / \tau$, where τ is the cell time constant set by the roller clamps.

Heat problem	Rig
Layer temperature	Water level in a syringe (cm above the zero mark)
Layer heat capacity	Syringe cross-section area

Heat problem	Rig
Thermal resistance between layers	IV tube with a roller clamp
Ambient temperature, left	Water level in the tank (bottle on a stand)
Ambient temperature, right	Height of the drain tube end
Time	Time, at its own scale (see section 5)

The main health check: keep the tank at a constant level, and after a while the syringe levels line up on a straight line from the tank to the drain. The program draws this line dashed.

2. Parts (for 5 syringes)

#	Part	Qty	Notes
1	Plastic bottle 0.5–1 L	1	Cut off the bottom; hangs neck down — this is the tank
2	Syringe 60 ml	5	Inner Ø ≈ 26.7 mm, 60 ml scale ≈ 10.7 cm. Plungers not needed
3	3-way stopcock (Luer)	5	Sold in pharmacies; one under each syringe
4	IV drip set	6	Only the tube and roller clamp are used; no drip chamber or needle
5	Roller clamp	—	Part of the drip set; it is the adjustable resistance
6	Cup	1	Under the drain

#	Part	Qty	Notes
7	Stand, rail, clips or clamps	2	Hold the tank (movable) and the drain tube end (fixed)
	Spirit level, ruler, tape		To set all scale zeros at the same height
	Food colouring		Levels are easier to see on video
	Phone on a tripod		Timelapse to compare with the program

There is one more drip set than syringes: tubes run tank → #1 → … → #5 → drain, 6 pieces in total, each with its own clamp.

3. Assembly

1. **Drip sets.** Cut off the drip chamber and the needle. Keep a ~30 cm piece of tube with the roller clamp in the middle. Make all pieces the same length.
2. **Syringes.** Remove the plungers. Mount the syringes on a rail vertically, nozzle down, evenly spaced. **All scale zero marks must be at exactly the same height** — check with a spirit level or a taut string. This matters most for accuracy.
3. **Stopcocks.** Insert each syringe nozzle into the side port of a stopcock. Connect the tubes to the straight ports: push them on tightly or use a Luer adapter. Turn the levers down — all three ports open.

4. **Tank.** Cut the bottom off the bottle, push the drip set spike into the cap, hang the bottle neck down on a stand. The water height in the bottle above the syringe zero is the “tank level”.

5. **Drain.** Fix the end of the tube from syringe #5 on the second stand at the required height above the cup. The height of the tube end above zero is the “drain level”.

6. **Filling.** Pour coloured water in through the tank with the drain clamped shut. Chase out bubbles: tap the tubes, wiggle them, tilt the stopcocks. A bubble in a tube changes its resistance sharply.

7. **Filming.** Place the phone square to the rail so that the scales of all syringes and the bottle are in frame. Start a timelapse (1–5 s interval).

4. Calibration

All pairs of neighbouring syringes must have the same **half-equalisation time t½** — the time for the level difference in a pair to halve. This is the **pair_half_time_s** parameter in the program (default 20 s, $\tau = 2 \cdot t_{½} / \ln 2 \approx 57.7$ s).

Poiseuille estimate: an open Ø3 mm × 30 cm tube gives $t_{½} \approx 3.0$ s — too fast to film. Slow it down to 15–30 s with the clamps.

The stopcock lever points at the closed port (on most models — check yours). Lever down, towards the missing port, means all three are open.

Procedure for pair i – (i+1)

1. Stopcock i: close the port towards the previous syringe (or the tank). Stopcock i+1: close the port towards the next syringe (or the drain). Now water flows only between i and i+1.

2. Fill syringe i with about 42 ml, empty syringe i+1 (through its stopcock).

3. Start a stopwatch. Note when the level difference has dropped to half the initial one.

4. If t½ is under 20 s, tighten the clamp; if over, loosen it. Repeat.

5. Do this for all pairs #1–#2 … #4–#5.

6. End tubes (tank–#1 and #5–drain): leave the syringe connected only to the tank (or drain). Its deviation from the tank level should halve in **2·t½** — the bottle is wide, so its level barely changes.

Rig check

Open all stopcocks, raise the tank to e.g. 8 cm, drain at 0. After 5–10 τ the levels should form a straight line. If the line “bends”, the tube at that spot has a different resistance: a bubble, an over-tightened clamp or a pinched tube.

5. How to run calculations

A real problem is converted into rig levels and seconds, the rig “computes”, and the answer is converted back. Three scales are needed.

Scale	Formula	Meaning
Length	$\Delta x = L / (N + 1) = L / 6$	A body of thickness L is split into N+1 intervals; tank and drain sit on the surfaces, syringes inside

Scale	Formula	Meaning
Time	$\tau_{body} = \Delta x^2 / a$; $M = \tau_{body} / \tau_{rig}$	a is the material's thermal diffusivity, $\tau_{rig} \approx 57.7$ s. 1 s on the rig = M s in reality
Temperature	$h = (T - T_{min}) / (T_{max} - T_{min}) \cdot 10$ cm	The temperature range is mapped onto the scale (10.7 cm). 1 cm of level = 5.6 ml

Material	a, m²/s	Material	a, m²/s
Concrete	$7 \cdot 10^{-7}$	Brick	$5 \cdot 10^{-7}$
Wood (pine, across grain)	$1 \cdot 10^{-7}$	Steel	$1.4 \cdot 10^{-5}$
Moist soil	$5 \cdot 10^{-7}$	Mineral wool	$1 \cdot 10^{-6}$

Procedure

1. Calibrate the rig (section 4).

2. Convert the problem into rig centimetres and seconds with the formulas above.

3. Initial state: close the stopcocks, fill each syringe with the required ml, open the stopcocks.

4. Boundaries: raise the tank to its height, fix the drain at its own.

5. Start the timelapse. If the ambient temperature changes (day/night, scheduled heating), move the tank by hand on schedule. This is exactly the kind of problem an analog integrator handles best.

6. Convert the readings back: ml → cm → °C, rig time × M → real time.

6. Example: a concrete wall freezing through

A 30 cm wall, +20 °C inside, and a sudden −20 °C frost outside. How does the cold move through the wall and when does it settle?

Quantity	Calculation	Value
Δx	0.30 m / 6	5 cm
τ_{body}	$0.05^2 / 7 \cdot 10^{-7}$	≈ 3571 s ≈ 1.0 h
M	3571 / 57.7	≈ 62: 1 rig min ≈ 1.0 h
Temperature	40 °C over 10 cm	1 cm = 4 °C; 0 cm = −20 °C, 10 cm = +20 °C
Start	Wall at +20 °C throughout	all syringes 10 cm (56 ml)
Tank (room)	+20 °C	10 cm

Quantity	Calculation	Value
Drain (outside)	−20 °C	0 cm

Reading the answer

Say after 3 rig minutes (≈ 3.1 h of real time) syringe #3 (middle of the wall) shows 43 ml (as the program gives). Then h = 43 / 5.6 ≈ 7.7 cm, and the temperature T = −20 + 7.7 · 4 ≈ 11 °C. After 10–15 rig minutes the levels form a straight line — that is the steady state; the wall will not freeze any further.

Limitations

• The body is uniform: layers of different materials need different t½ per pair (the program cannot do that yet).

• The tank tube equals an ordinary layer, i.e. the wall is in perfect contact with the environment. Heat exchange with air (1/α) can be

modelled by tightening the tank and drain clamps more.

• 5 layers give a coarse but usable engineering profile: error of a few percent. More syringes — more accurate.

7. Using the program

Key	Action	Key	Action
↑ / ↓	Tank up / down 1 cm	S	Save readings to CSV
Space	Pause	F1	This manual
+ / −	Time speed ×2 / ÷2	← / →	Manual pages
R	Reset	O	Open the PDF in an external viewer
L	Language: English / Russian	[/]	Inflow □0.5 ml/min

Mouse (and touch on Steam Deck)

Action	Effect
Drag the bottle	Tank up / down; the level is shown above the bottle
Drag the drain tube end	Changes the drain level
Click or drag inside a syringe	The water level jumps to the cursor — this is how you set the initial state
Wheel over tank, drain or syringe	Fine adjustment ±0.1 cm
Top buttons	Pause/Run, − / + speed, Reset, − / + inflow, Save CSV, Manual, language

Initial state with the mouse: press “Pause”, set the syringe levels and the tank, then “Run”. Handy for reproducing the problems in sections 6 and 8.

Parameters (Inspector, SyringeRig node)

Parameter	Meaning
pair_half_time_s	t½ of a syringe pair — from calibration (section 4)
syringe_volume_ml, syringe_inner_d_mm	Volume and inner diameter of your syringes — they set the scale height
tube_inner_d_mm, tube_length_cm	Drip tubes — for the open-tube t½ estimate
tank_level_cm, outlet_level_cm	Tank and drain levels at start
start_level_cm	Initial level in all syringes
record_interval_s	How often to write a CSV row
inflow_cm_per_min, inflow_decay_min	Inflow into each syringe (heat source) and its decay; counted from reset
drops_per_ml	Drops per ml of the inflow drip set (usually 20)

Comparing with the rig

The CSV is saved to user://lukyanov_levels.csv (the path is shown on screen): t_s, tank_cm, h1_cm … h5_cm. Set the same t½, tank and drain in the program as on the rig, start together with the timelapse, save the CSV and compare with levels read from the video (ml ÷ 5.6 = cm). Differences point to bubbles, unequal clamps or scale zeros at different heights.

8. History: how the integrator helped builders

The problem

In the late 1920s engineer Vladimir Sergeyevich Lukyanov worked on railway construction (including the Troitsk–Orsk line). Concrete structures cracked: setting concrete releases heat while outside there is frost or heat, so temperature differences and stresses build up inside.

To know where and when cracks appear you need the temperature inside the concrete day by day. That is the heat equation with changing conditions: weather, heat of setting, formwork insulation. It could not be solved exactly on paper, and approximate hand calculations took weeks and were crude.

The solution

In 1936 Lukyanov built the hydraulic integrator — exactly the principle of this rig: vessels are concrete layers, tubes are heat conduction, water level is temperature. The engineer set initial levels, moved the tank on schedule (“the weather outside”) and took readings. A problem that used to take weeks was solved in hours.

The main advantage was trying variants quickly: “what if we insulate”, “what if we pour in another month”, “what if we heat it” — and picking the best before work began.

Where it was used

Field	What was calculated
Winter concreting	How long to keep formwork and insulation so concrete neither freezes nor cracks
Permafrost	How an embankment, building or foundation changes the permafrost below: will the ground thaw and settle. Northern railways, towns, mines
Mining and geology	Freezing and thawing of rock around mines, subsurface thermal regime
Metallurgy and more	Cooling of ingots; groundwater seepage, diffusion — anything that reduces to the same equation

Development

Years	Event
1936	First one-dimensional integrator (like this rig)
1941	Two-dimensional integrator: a whole dam or foundation cross-section, not a single line
1950s	Series production at the Ryazan calculating-machine plant; integrators worked in dozens of institutes (by some accounts over a hundred organisations), including abroad
until 1980s	In use until digital computers replaced them: for these problems the hydraulic machine long stayed clearer and more reliable than early computers

Two machines are preserved at the Polytechnic Museum (Moscow). Exact dates and numbers of machines differ between sources.

Problems by field: from a real situation to the rig settings

Every problem follows the procedure of section 5: $\Delta x = L / 6$, $\tau_{body} = \Delta x^2 / a$, $M = \tau_{body} / 57.7$ s, temperatures are mapped to 0–10 cm. Rig times and results are computed with the same model as the program.

Boundary rule. The drain can only take water away, so put the **cold** (lower) boundary on the drain and the **warm** one on the tank. The tank bottle can both give and take water.

Problem 1. Concreting in winter

Problem. A 40 cm monolithic wall is cast with concrete at +15 °C (heated mix). The formwork is removed, it is −15 °C outside on both sides. How many hours until the concrete starts freezing near the surface and in the middle? Is a heated enclosure needed?

Quantity	Calculation	On the rig
Layers	$\Delta x = 0.40\ \text{m} / 6$	6.7 cm per syringe
Time	$\tau = 0.067^2 / 7 \cdot 10^{-7} = 1.8\ \text{h}$	M ≈ 110: 1 min ≈ 1.8 h
Temperature	−15…+15 °C over 10 cm	1 cm = 3.0 °C; 0 °C = 5.0 cm
Start	All concrete at +15 °C	all syringes 10.0 cm (56 ml)
Tank and drain	Both faces −15 °C	tank 0 cm, drain 0 cm
Find	When a layer drops below 0 °C	level falls to 5.0 cm (28 ml)

Solution. Syringe #1 (the layer 6.7 cm from the face) falls to 5.0 cm after 1 min 3 s on the rig — that is ≈ **1.9 h**. The middle (#3) — after 3 min 16 s, ≈ **6.0 h**.

Concrete must not freeze before it reaches its critical strength, which usually takes a day or more. So without protection the wall freezes too early.

With a heated enclosure (air around +5 °C): tank and drain at 6.7 cm. Levels never fall below that mark, so no layer goes below zero — the enclosure solves the problem. Trying out variants like this was the integrator's main job.

Problem 2. A building on permafrost

Problem. A heated building stands directly on the ground: +12 °C under the floor. Permafrost is −5 °C; at 6 m depth its temperature does not change. How fast and how deep will the ground under the building thaw?

Quantity	Calculation	On the rig
Layers	$\Delta x = 6\ \text{m} / 6$	1 m: #1 — depth 1 m, …
Time	$\tau = 1^2 / 5 \cdot 10^{-7} = 23$ days	M ≈ 34657: 1 min ≈ 24 days
Temperature	−5…+12 °C over 10 cm	1 cm = 1.7 °C; 0 °C = 2.94 cm
Start	All ground at −5 °C	all syringes 0 cm (empty)
Tank / drain	Floor +12 °C / at 6 m −5 °C	tank 10 cm, drain 0 cm
Find	When a layer thaws (above 0 °C)	level rises above 2.94 cm

Solution. The thaw reaches depth: 1 m — 10 days; 2 m — 42 days; 3 m — 3 months; 4 m — 7 months.

In steady state the levels form a straight line from 10 cm to 0, and the 0 °C boundary stops at a depth of ≈ **4.2 m**. Thawed ground settles — the building will deform.

With a ventilated crawl space: about −2 °C under the floor on average over the year, tank at 1.76 cm — below the 0 °C mark. No thaw. That is why northern houses stand on piles with a ventilated space underneath.

The model ignores the latent heat of melting ice, so real thawing is slower — for design purposes this is a conservative estimate.

Problem 3. An ice wall around a mine shaft

Problem. A mine shaft is sunk through water-bearing ground at +8 °C. Boreholes around the shaft carry −25 °C brine so the ground freezes and water cannot break through. How many days until the ice wall is 1 m thick? How thick will it eventually get?

Quantity	Calculation	On the rig
Layers	$\Delta x = 3\ \text{m} / 6$	0.5 m per syringe, #5 next to the boreholes
Time	$\tau = 0.5^2 / 5\cdot10^{-7} = 5.8$ days	M ≈ 8664: 1 min ≈ 6 days
Temperature	−25…+8 °C over 10 cm	1 cm = 3.3 °C; 0 °C = 7.58 cm
Start	All ground at +8 °C	all syringes 10 cm (56 ml)
Tank / drain	Ground +8 °C / brine −25 °C	tank 10 cm, drain 0 cm
Find	When a layer freezes	level below 7.58 cm

Solution. Freezing from the boreholes: 0.5 m — 44.4 h; 1.0 m — 8 days; 1.5 m — 19 days; 2.0 m — 38 days.

A 1 m wall is the moment syringe #4 drops below 7.58 cm. The limit thickness is ≈ **2.3 m** (where the 0–10 cm straight line crosses the 0 °C mark).

Freezing water releases a lot of heat, which the model ignores. The real wall grows more slowly, so these times are a lower bound.

Problem 4. Quenching a steel slab

Problem. A 30 cm slab heated to 800 °C is dropped into 20 °C water. When does the middle cool below 200 °C so the slab can be taken out?

Quantity	Calculation	On the rig
Layers	$\Delta x = 0.30\ \text{m} / 6$	5.0 cm per syringe
Time	$\tau = 0.05^2 / 1.4 \cdot 10^{-5} = 179\ \text{s}$	M ≈ 3.1: 1 min ≈ 3.1 min
Temperature	20…820 °C over 10 cm	1 cm = 80 °C; 200 °C = 2.25 cm
Start	Slab at 800 °C	all syringes 9.75 cm (55 ml)
Tank and drain	Both faces in 20 °C water	tank 0 cm, drain 0 cm
Find	Middle below 200 °C	syringe #3 below 2.25 cm

Solution. The layer 5 cm from the face (#1) cools to 200 °C after 3 min 34 s on the rig — ≈ **11.0 min**; the middle (#3) — after 6 min 2 s, ≈ **18.7 min**.

The time scale here is about 3: the rig runs only three times faster than reality. That is fine — the integrator is chosen for clarity and for trying variants, not for speed.

The surface is assumed to be at water temperature right away; in reality a vapour blanket first forms around hot steel, and cooling is slower for the first minutes.

Problem 5. Heat of setting in mass concrete

Problem. A 1.2 m foundation slab is poured with concrete at +10 °C, air also +10 °C. Setting cement releases heat: with no losses the concrete would warm up by 40 °C, and the heat release decays with a time constant of ≈ 3 days. How hot does the middle get? And what is the thickest slab that can be poured in one go while the core-to-surface difference stays within 20 °C (the usual anti-cracking guideline)?

Quantity	Calculation	On the rig
Layers	$\Delta x = 1.2\ \text{m} / 6$	20 cm per syringe
Time	$\tau = 0.2^2 / 7 \cdot 10^{-7} = 15.9$ h	M ≈ 990: 1 min ≈ 16.5 h
Temperature	+10…+70 °C over 10 cm	1 cm = 6 °C
Start, tank, drain	Everything +10 °C	syringes, tank and drain 0 cm

Quantity	Calculation	On the rig
Inflow	40 °C = 6.67 cm over 3 days = 4.4 rig min	8.6 ml/min (171 drops/min) per syringe, decay T = 4.4 min
Find	Peak in the middle	highest level of syringe #3

Solution. The middle (#3) peaks at 2.67 cm after 4 min 5 s on the rig — that is ≈ **26 °C** after ≈ **3 days**. Core-to-surface difference ≈ **16 °C** — below 20 °C: this slab can be poured in one go.

In the program: in the Inspector of the SyringeRig node set tank_level_cm = 0, start_level_cm = 0, inflow_cm_per_min = 1.53 (= 8.6 ml/min), inflow_decay_min = 4.4 and run. **On the rig:** a drip from a separate bottle above each syringe; reduce it on schedule: 0 min — 171 drops/min; 4 min — 63 drops/min; 9 min — 23 drops/min; 13 min — 9 drops/min. Water flowing into the tank raises its level — scoop it out to keep the tank at zero.

Trying thicknesses

The same calculation for other thicknesses (each has its own time scale, hence its own inflow and decay). Over the limit — in bold:

Thickness	Peak in the middle	Difference	Inflow on the rig
0.6 m	17 °C after 29.9 h	7 °C	2.1 ml/min
1.0 m	23 °C after 2 days	13 °C	5.9 ml/min
1.2 m	26 °C after 3 days	16 °C	8.6 ml/min
1.5 m	30 °C after 4 days	20 °C	13.4 ml/min
2.0 m	35 °C after 5 days	**25 °C**	23.8 ml/min

The limit is ≈ **1.5 m**: thicker slabs must be poured in lifts or cooled. This is how the integrator helped choose lift thickness and curing regime.

Thick slabs need a strong inflow — more than a drip set can give. Then tighten the clamps between the syringes (larger t½): the rig runs slower and the inflow

drops by the same factor.

The surface is assumed to be at air temperature and concrete properties constant; with insulated formwork the surface is warmer and the difference smaller. A real project needs heat-release data for the specific cement.

What else you can reproduce on this rig

• **Freezing of a wall or ground** under changing weather — already possible: move the tank following the outdoor temperature (example — section 6).

• **Heat of concrete setting** — a heat source inside each layer: water dripping into each syringe from a separate drip (problem 5). Here the IV drip set is used for its original purpose.